

Course Registration & Enrollment System

Web Engineering (CSC-324) – Assignment 3

Instructor: Syed Nouman Ali Shah

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Student Name: Saad Mugha;

Roll Number: F2023-009

Program: BS Computer Science

Semester: 5

Executive Summary

This report documents the development of a comprehensive Course Registration and Enrollment System built using Spring Boot for the backend and React for the frontend. The system implements a complete domain model for managing students, courses, and registrations with robust business rule validation and RESTful API architecture.

Technology Stack:

Spring Boot 3.x Spring Data JPA MySQL Database React 18 React Router Axios RESTful APIs

Part 1: Domain Model & JPA Entities

1.1 Entity Design & Relationships

The system implements three core entities with proper JPA mappings and bidirectional relationships:

Student Entity

Attribute	Type	Constraints
id	Long	Primary Key, Auto-generated

rollNo	String	Unique, Not Null
name	String	Not Null
email	String	Unique, Valid Email Format
program	String	Not Null (e.g., BSCS, BSE)
semester	Integer	Min: 1, Max: 8

Course Entity

Attribute	Type	Constraints
id	Long	Primary Key, Auto-generated
code	String	Unique, Not Null (e.g., CSC-324)
title	String	Not Null
creditHours	Integer	Min: 1, Max: 6
maxSeats	Integer	Min: 1
prerequisite	Course	Self-reference (Optional)

Registration Entity

Attribute	Type	Constraints
id	Long	Primary Key, Auto-generated
student	Student	Many-to-One, Not Null
course	Course	Many-to-One, Not Null
semester	String	Not Null (e.g., Fall 2025)
status	Enum	REGISTERED, COMPLETED, DROPPED
grade	String	Optional (A, A-, B+, etc.)
registrationDate	LocalDate	Auto-set on creation

1.2 JPA Annotations & Validations

Key JPA annotations implemented:

- **@Entity** - Marks classes as JPA entities
- **@Id & @GeneratedValue** - Primary key with auto-increment strategy
- **@ManyToOne & @OneToMany** - Bidirectional relationships
- **@JoinColumn** - Foreign key column definitions
- **@NotNull, @NotBlank** - Bean validation
- **@Email** - Email format validation
- **@Min, @Max** - Numeric range validation
- **@Enumerated** - Enum type mapping for status

1.3 Repository Layer

Custom query methods implemented in repositories:

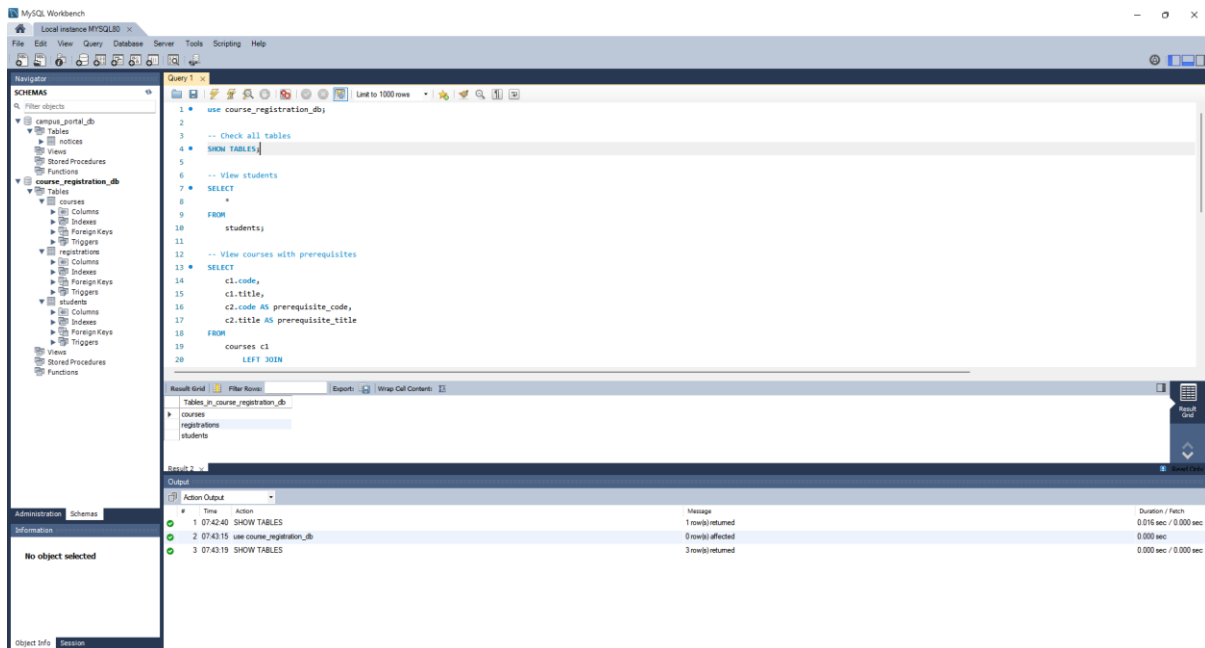
```
// StudentRepository Optional<Student> findByRollNo(String rollNo);
// CourseRepository Optional<Course> findByCode(String code);
// RegistrationRepository List<Registration> findByStudentId(Long studentId);
// List<Registration> findByCourseId(Long courseId);
```

```
List<Registration> findBySemester(String semester);
List<Registration> findByStudentIdAndStatus(Long studentId, RegistrationStatus status);
```

1.4 Sample Data Seeding

A **DataLoader** component seeds the database with initial data on application startup:

- 5 Students with diverse programs (BSCS, BSE, BSDS)
- 5 Courses with prerequisite relationships (CSC-201 → CSC-324)
- 5 Sample registrations with different statuses



Part 2: RESTful API Design & Business Rules

2.1 Student API Endpoints

Method	Endpoint	Description	Status Code
GET	/api/students	Retrieve all students	200 OK
GET	/api/students/{id}	Get single student by ID	200 OK / 404 Not Found
POST	/api/students	Create new student	201 Created

PUT	/api/students/{id}	Update existing student	200 OK
DELETE	/api/students/{id}	Delete student	204 No Content
GET	/api/students/{id}/registrations	Get student's registrations	200 OK

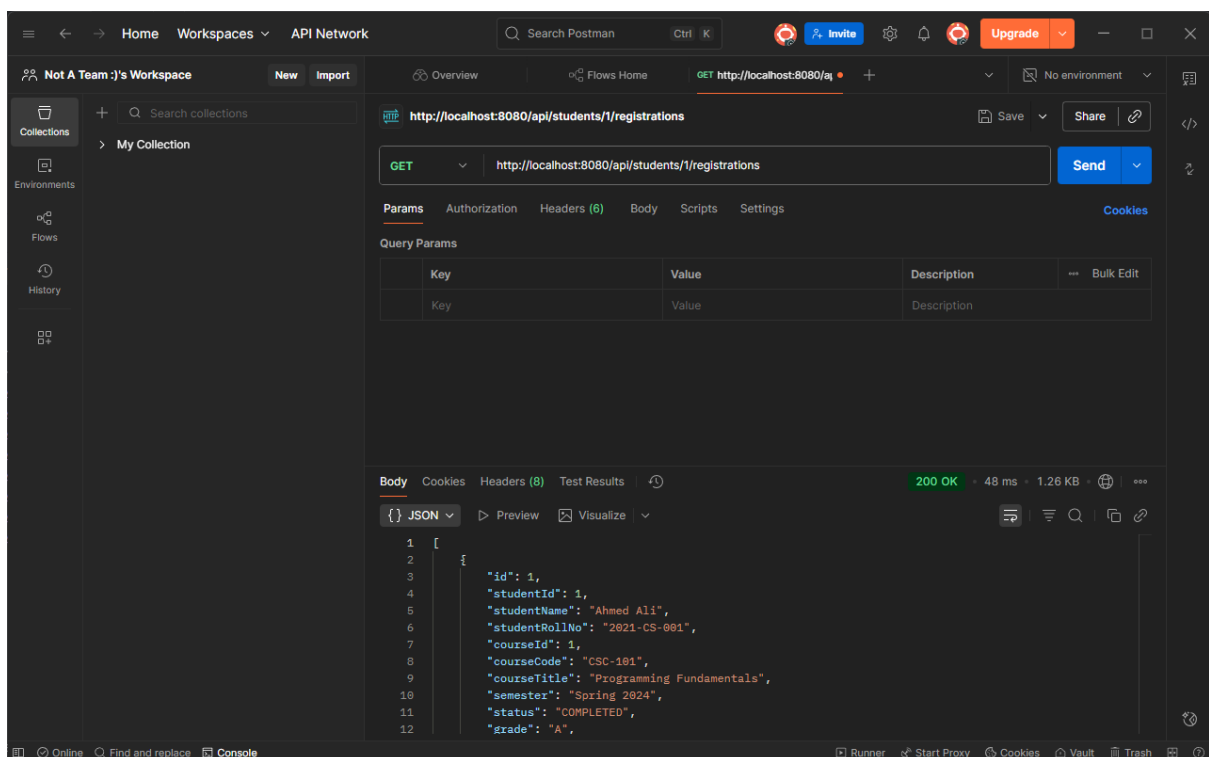
2.2 Course API Endpoints

Method	Endpoint	Description	Status Code
GET	/api/courses	Retrieve all courses	200 OK
GET	/api/courses/{id}	Get single course by ID	200 OK / 404 Not Found
POST	/api/courses	Create new course	201 Created
PUT	/api/courses/{id}	Update existing course	200 OK
DELETE	/api/courses/{id}	Delete course	204 No Content
GET	/api/courses/{id}/registrations	Get course registrations	200 OK

2.3 Registration API Endpoints

Method	Endpoint	Description	Status Code
POST	/api/registrations	Register student for course	201 Created / 400 Bad Request

GET	/api/registrations	Get all registrations	200 OK
GET	/api/registrations/{id}	Get registration by ID	200 OK / 404 Not Found
PATCH	/api/registrations/{id}	Update registration status/grade	200 OK
DELETE	/api/registrations/{id}	Delete registration	204 No Content



2.4 Business Rules Implementation

The system enforces three critical business rules in the **RegistrationService** layer:

Rule 1: Maximum Course Limit

A student cannot register for more than 5 active courses (status = REGISTERED) in the same semester.

Implementation: Query count of active registrations before allowing new registration.

```

if (activeRegistrations >= 5)
{ throw new BusinessException( "Student cannot register for more than 5
courses in the same semester" ); }

```

Rule 2: Maximum Seat Capacity

A course cannot exceed its maximum seat capacity for a given semester.

Implementation: Count current registrations and compare with maxSeats.

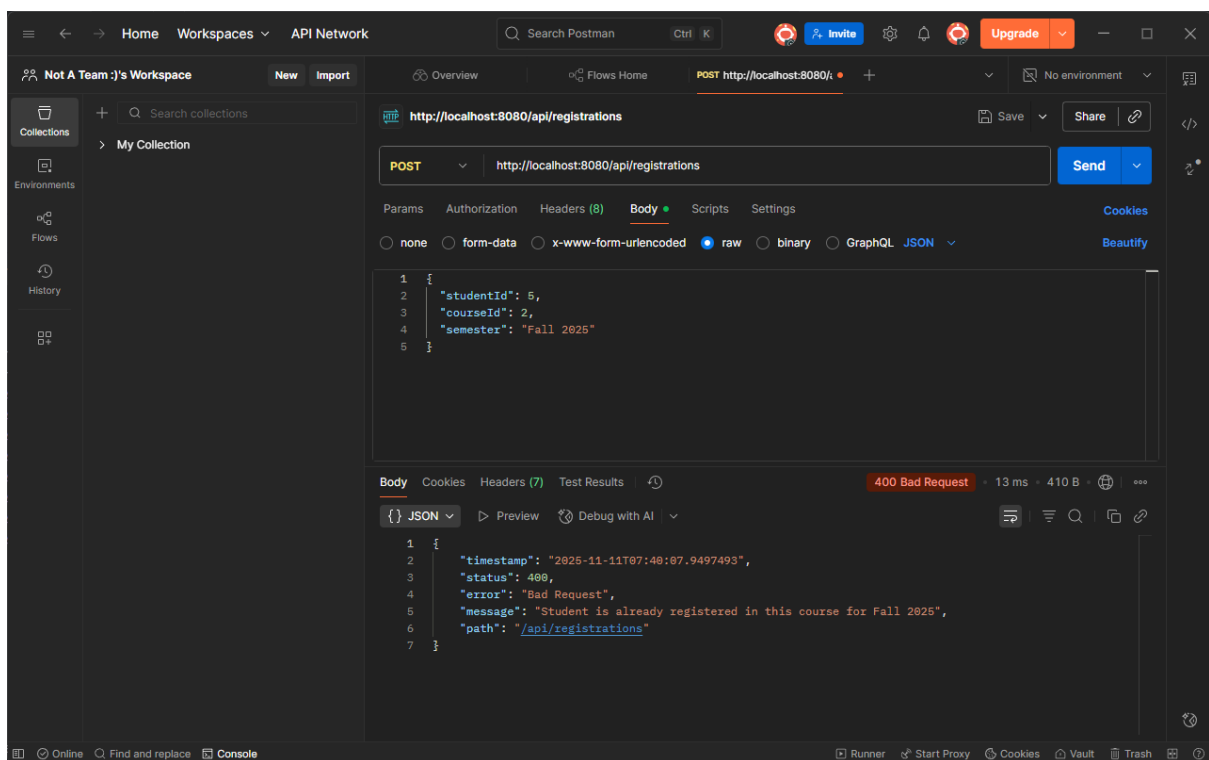
```
if (currentEnrollment >= course.getMaxSeats())  
{ throw new BusinessException( "Course has reached maximum capacity" ); }
```

Rule 3: Prerequisite Validation

If a course has a prerequisite, the student must have completed it (status = COMPLETED) before registering.

Implementation: Check for completed prerequisite registration.

```
if (course.getPrerequisite() != null)  
{ boolean hasCompleted = registrationRepository.existsByStudentAndCourseAndStatus( student,  
course.getPrerequisite(), RegistrationStatus.COMPLETED );  
  
if (!hasCompleted)  
{ throw new BusinessException( "Student must complete prerequisite: " +  
course.getPrerequisite().getCode() ); } }
```



2.5 Exception Handling Architecture

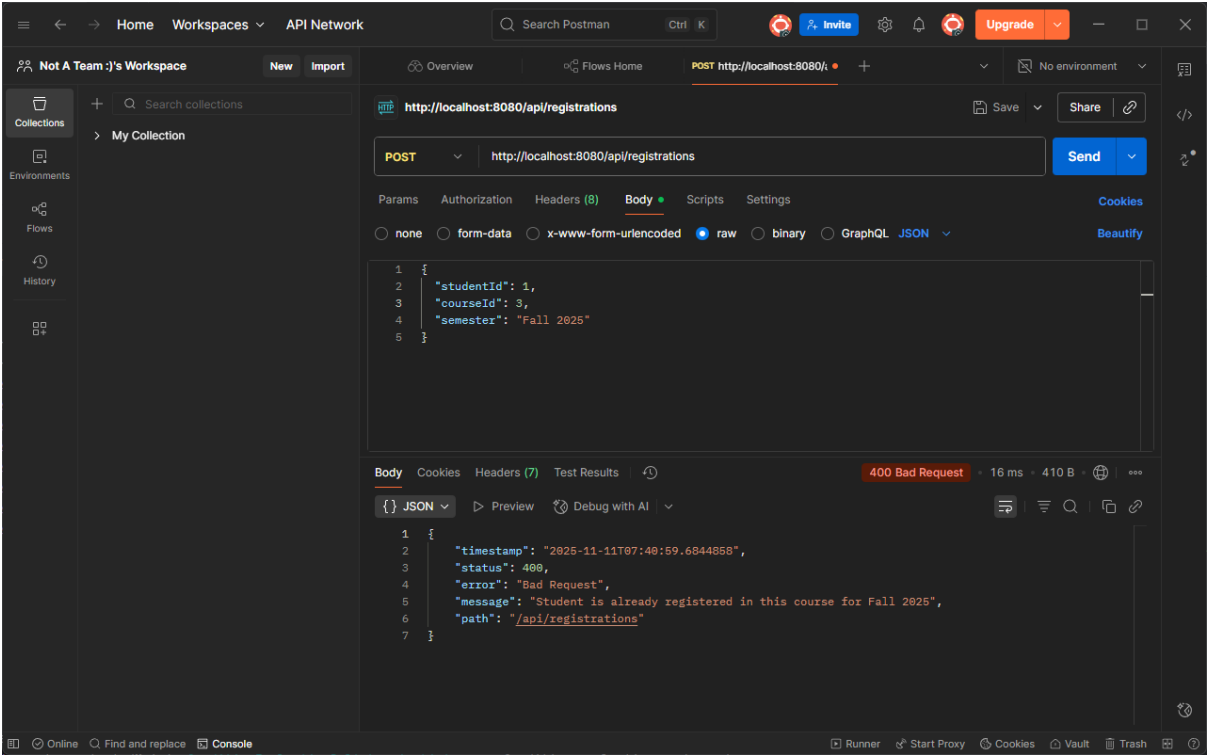
Custom exceptions with `@RestControllerAdvice` for centralized error handling:

- **StudentNotFoundException** - Returns 404 with error message
- **CourseNotFoundException** - Returns 404 with error message

- **RegistrationNotFoundException** - Returns 404 with error message
- **BusinessRuleViolationException** - Returns 400 with violation details

All exceptions return consistent JSON format:

```
{ "timestamp": "2025-11-11T10:30:00", "status": 400, "error": "Bad Request", "message": "Student cannot register for more than 5 courses", "path": "/api/registrations" }
```



Part 3: React Frontend Implementation

3.1 Application Architecture

The React frontend follows a component-based architecture with clean separation of concerns:

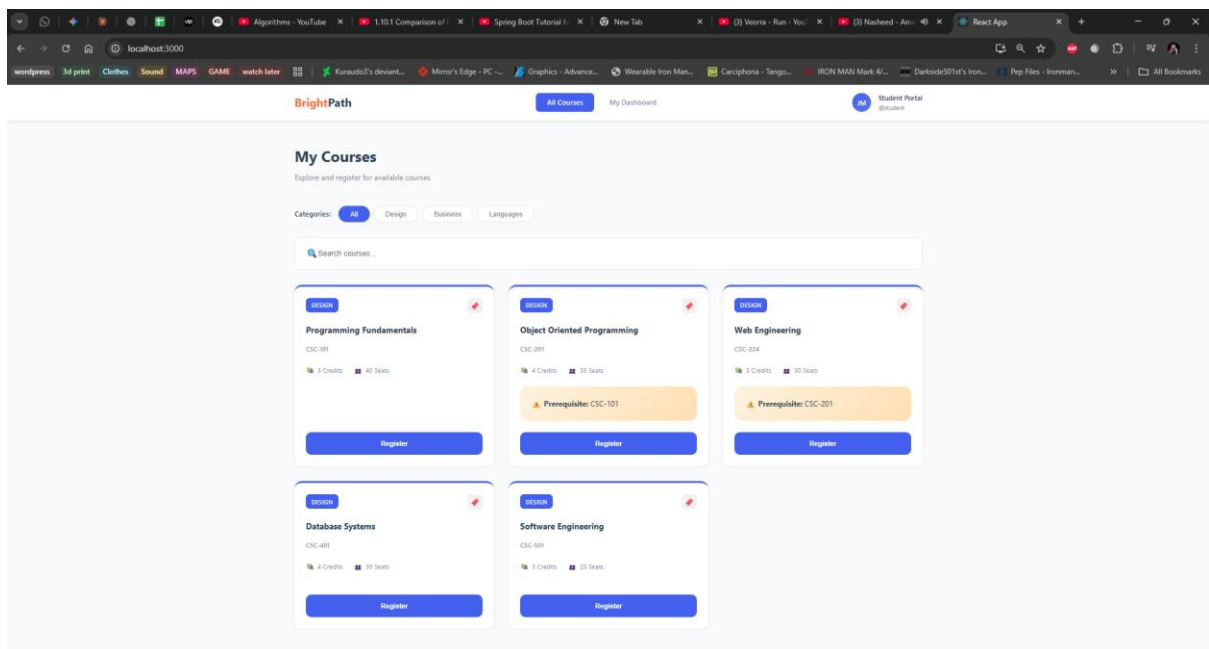
Component	Responsibility	Key Features
App.js	Main application shell	Routing, navigation, layout structure
CourseList.js	Display available courses	Search, filter, registration trigger

RegistrationForm.js	Course registration modal	Student selection, semester choice, submission
StudentDashboard.js	Student registration view	Student selection, status filter, data table
api/api.js	API integration layer	Axios configuration, endpoint definitions

3.2 Course Listing & Search Feature

The **CourseList** component provides:

- Fetches all courses from GET /api/courses on component mount
- Real-time search filtering by course code or title
- Category-based filtering (Design, Business, Languages)
- Responsive card-based layout with hover effects
- Displays course details: code, title, credits, max seats, prerequisites
- Visual prerequisite warnings with highlighted badges

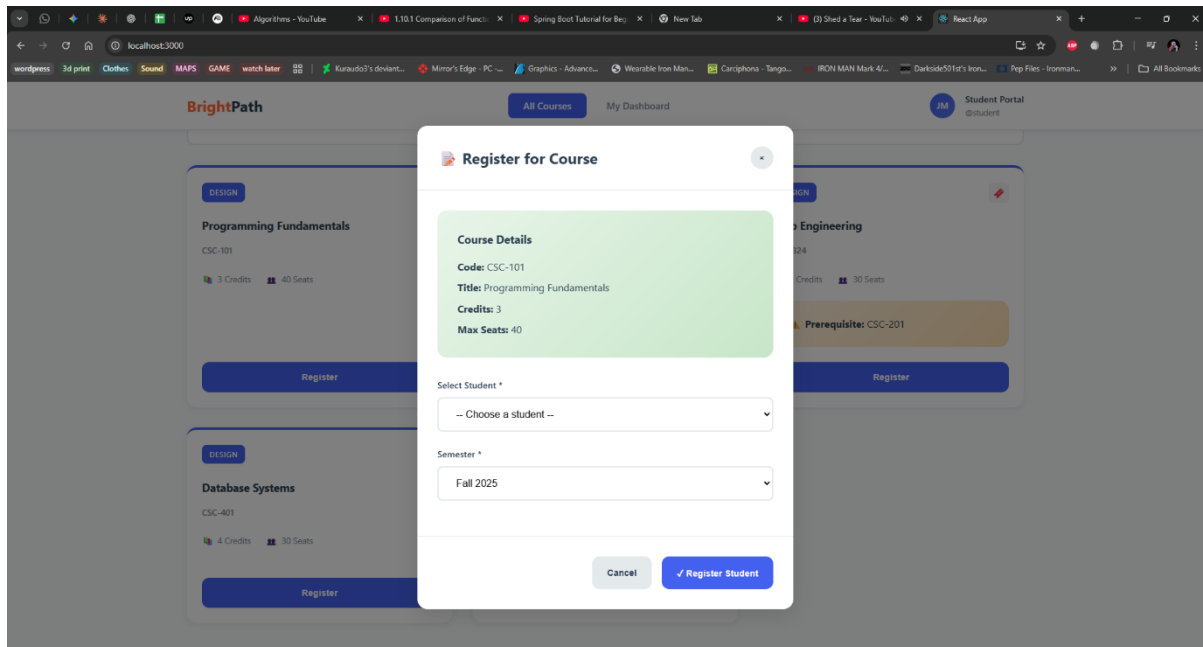


3.3 Registration Form & Modal

The **RegistrationForm** component implements:

- Modal overlay with course information display
- Student dropdown populated from GET /api/students
- Semester selection dropdown (Fall/Spring/Summer)
- Form validation before submission

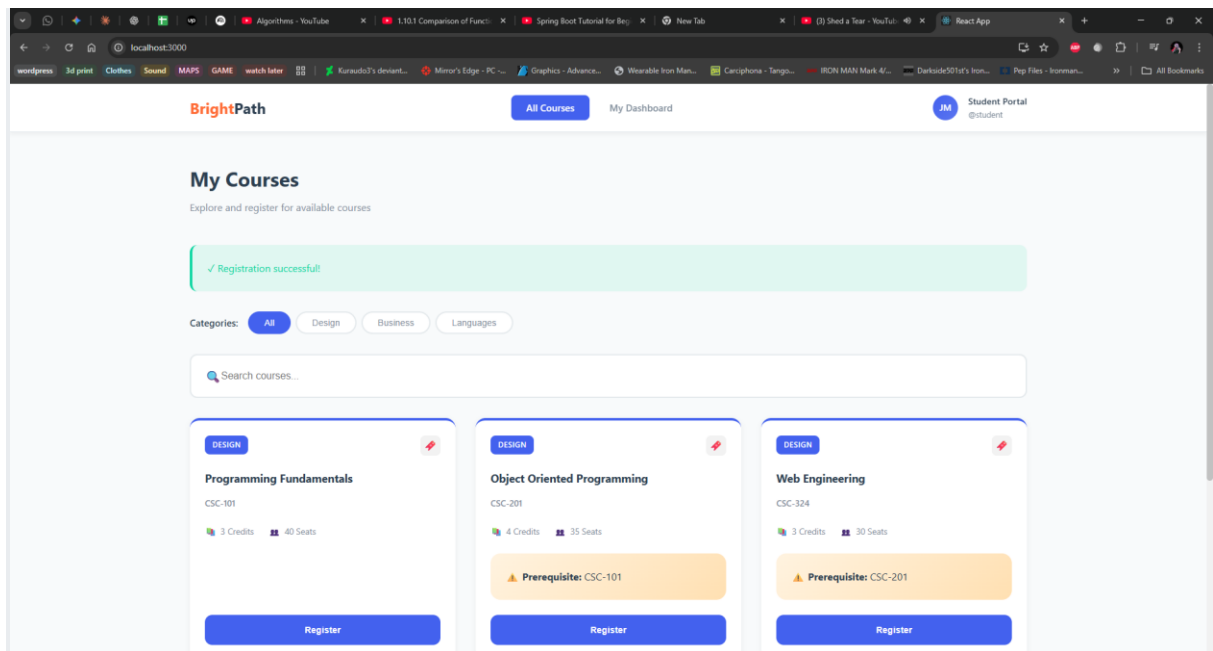
- POST request to `/api/registrations` with proper error handling
- Success alerts with green notification
- Error alerts showing business rule violations
- Loading state during API call with disabled submit button



✓ Successful Registration Flow

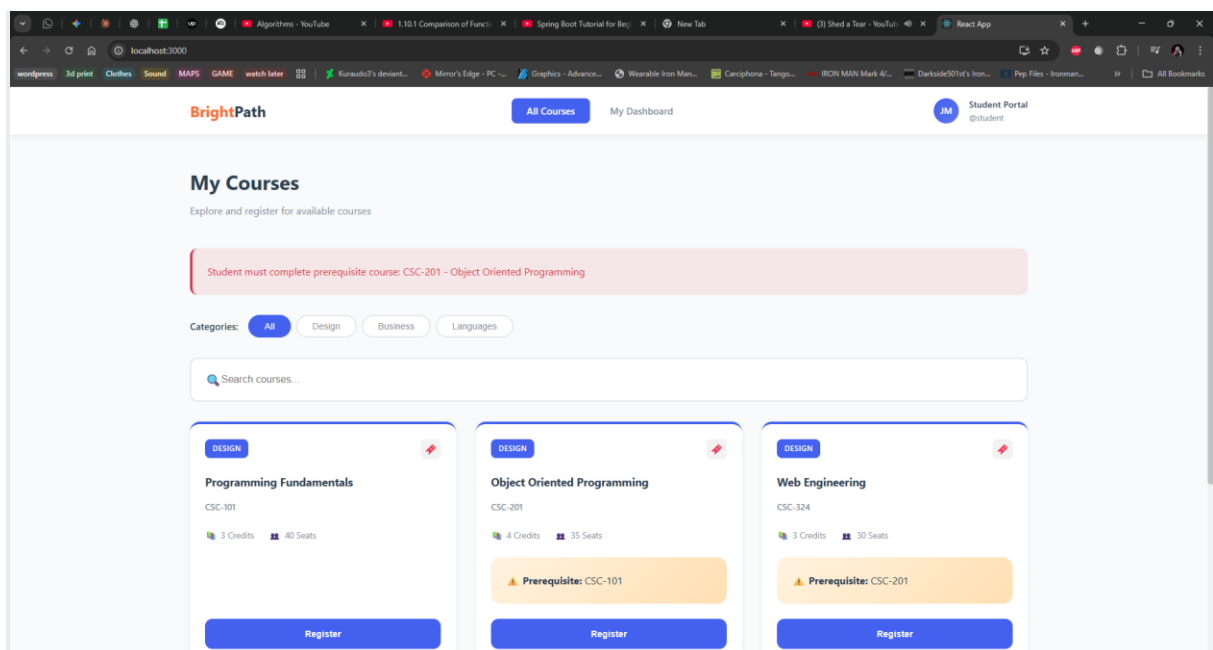
When registration succeeds, the system:

1. Displays success alert: "✓ Registration successful!"
2. Closes the modal automatically
3. Refreshes the course list to update available seats
4. Auto-dismisses alert after 5 seconds



⚠ Error Handling Example

When business rules are violated:

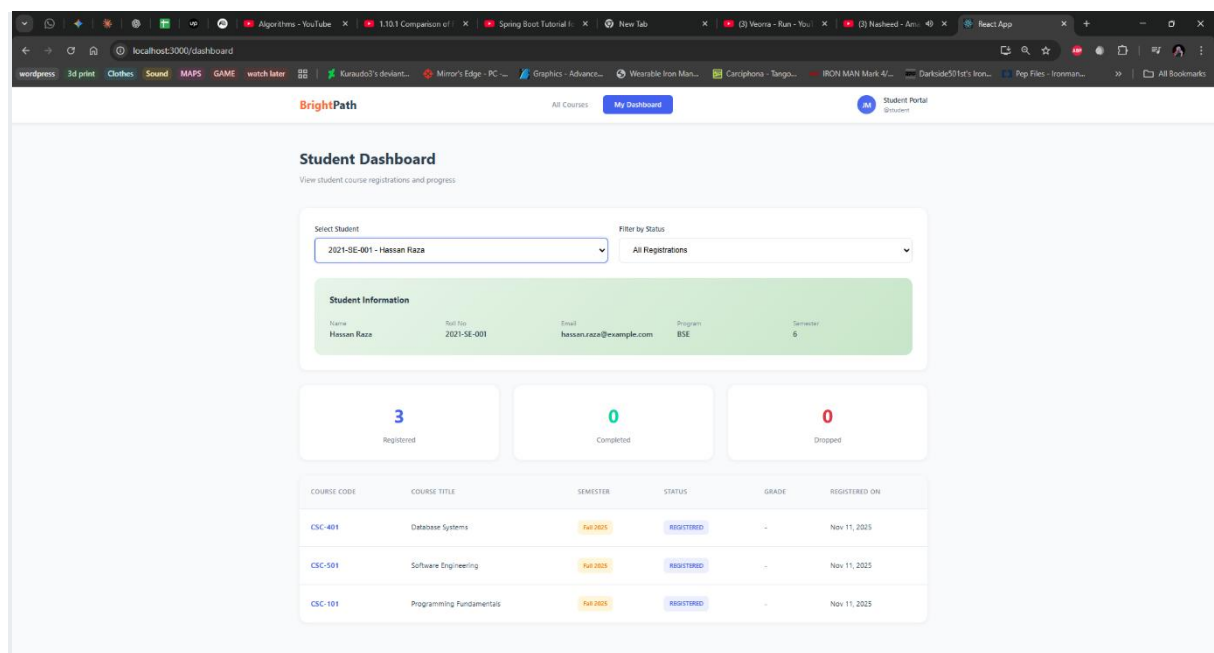


3.4 Student Dashboard

The **StudentDashboard** component provides comprehensive student view:

Features:

- Student dropdown for selection
- Displays complete student information (name, roll no, email, program, semester)
- Statistics cards showing counts: Registered, Completed, Dropped courses
- Status filter to view specific registration types
- Data table with sortable columns: Course Code, Title, Semester, Status, Grade, Date
- Color-coded status badges (Blue: Registered, Green: Completed, Red: Dropped)
- Empty state message when no student is selected
- Loading indicators during data fetch



3.5 UI/UX Design Highlights

The interface incorporates modern design principles:

Aspect	Implementation
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Color Scheme	Primary Blue (#4361EE), Accent Orange (#FF6B35), Success Green (#06D6A0)
Typography	System font stack with proper hierarchy and readable sizes
Spacing	Consistent spacing system using CSS variables (--space-sm, --space-md, etc.)
Components	Cards with subtle shadows, rounded corners, hover effects
Responsiveness	Mobile-first grid layouts, adaptive navigation
Accessibility	Semantic HTML, proper labels, keyboard navigation support

Testing & Validation

4.1 Backend API Testing (Postman)

All endpoints were thoroughly tested using Postman:

Test Case	Expected Result	Status
GET /api/students	Returns all 5 students with 200 OK	✓ Pass
GET /api/courses	Returns all 5 courses with prerequisites	✓ Pass
POST /api/registrations (valid)	Creates registration, returns 201	✓ Pass
POST /api/registrations (duplicate)	Returns 400 with error message	✓ Pass

POST /api/registrations (no prerequisite)	Returns 400 "Must complete prerequisite"	✓ Pass
POST /api/registrations (max courses exceeded)	Returns 400 "Cannot register more than 5"	✓ Pass
POST /api/registrations (course full)	Returns 400 "Course at maximum capacity"	✓ Pass
GET /api/students/{id}/registrations	Returns student's registrations	✓ Pass
PATCH /api/registrations/{id}	Updates status/grade, returns 200	✓ Pass
DELETE /api/registrations/{id}	Deletes registration, returns 204	✓ Pass

4.2 Frontend Integration Testing

End-to-end testing scenarios verified:

Scenario 1: Successful Course Registration

1. Navigate to Course List page
2. Click "Register" on a course
3. Select student from dropdown
4. Choose semester
5. Click "Register Student"
6. **Result:** Success message displayed, modal closes

Scenario 2: Prerequisite Validation

1. Attempt to register a student in CSC-324 (Web Engineering)
2. Student has NOT completed CSC-201 (Data Structures)
3. **Result:** Error message "Student must complete prerequisite: CSC-201"

Scenario 3: Maximum Course Limit

1. Student already registered in 5 active courses
2. Attempt to register for 6th course
3. **Result:** Error message "Cannot register for more than 5 courses"

Scenario 4: Student Dashboard Navigation

- 1. Click "My Dashboard" in navigation
- 2. Select student from dropdown
- 3. Student info and statistics display
- 4. Registrations table populates with data
- 5. Apply status filter (e.g., "COMPLETED")
- 6. **Result:** Table filters to show only completed courses

4.3 Cross-Browser Compatibility

Browser	Version	Status
Google Chrome	120+	✓ Fully Compatible
Mozilla Firefox	121+	✓ Fully Compatible
Microsoft Edge	120+	✓ Fully Compatible
Safari	17+	✓ Fully Compatible

Technical Implementation Details

5.1 Project Structure

Backend Structure:

```
course-registration-frontend/  
├── public/  
├── src/  
│   ├── api/  
│   │   └── api.js           # Centralized API calls  
│   ├── components/  
│   │   ├── CourseList.js   # Displays list of available courses with filters  
│   │   ├── RegistrationForm.js # Modal/form for course registration  
│   │   └── StudentDashboard.js # Displays registered courses for the student  
│   ├── App.js              # Main app file handling routes  
│   ├── App.css              # Global styles  
│   ├── index.js             # Entry point  
└── package.json
```

Frontend Structure:

```
course-registration-frontend/
```

```

├── public/
├── src/
│   ├── api/
│   │   └── api.js                # Centralized API calls (Axios or Fetch)
│   ├── components/
│   │   ├── CourseList.js        # Displays available courses with filters
│   │   ├── RegistrationForm.js  # Handles course registration modal/form
│   │   └── StudentDashboard.js  # Shows student's registered courses
│   ├── App.js                   # Main app component with routing
│   ├── App.css                  # Global styling
│   ├── index.js                 # React entry point
│   ├── index.css                # Global base styles
│   └── package.json

```

5.2 Database Configuration

MySQL Database Setup:

```

# Database: course_registration_db
# Tables: student, course, registration
# Relationships:
#   - registration.student_id → student.id
#   - registration.course_id → course.id
#   - course.prerequisite_id → course.id (self-reference)

```

Application Properties:

```

spring.datasource.url=jdbc:mysql://localhost:3306/course_registration_db
spring.datasource.username=root spring.datasource.password=[password]
spring.jpa.hibernate.ddl-auto=update spring.jpa.show-sql=true
spring.jpa.properties.hibernate.format_sql=true

```

5.3 CORS Configuration

Enabled cross-origin requests for React frontend:

```

@CrossOrigin(origins = "http://localhost:3000") @RestController
@RequestMapping("/api/students") public class StudentController { // Controller methods }

```

5.4 Data Transfer Objects (DTOs)

Used DTOs to decouple API contracts from entity models:

```

// RegistrationRequestDTO.java public class RegistrationRequestDTO
{ private Long studentId; private Long courseId; private String semester; }

// RegistrationResponseDTO.java public class RegistrationResponseDTO
{ private Long id; private String studentName;

private String courseCode;
private String courseTitle;
private String semester;
private RegistrationStatus status;
private String grade;
private LocalDate registrationDate; }

```


Key Features & Achievements

6.1 Core Features Implemented

- **Complete CRUD Operations** - All entities support Create, Read, Update, Delete
- **Business Rule Validation** - Three comprehensive rules enforced at service layer
- **RESTful API Design** - Proper HTTP methods, status codes, and JSON responses
- **Exception Handling** - Centralized error handling with consistent JSON format
- **Data Validation** - Bean validation with @NotNull, @Email, @Min, @Max annotations
- **Bidirectional Relationships** - Proper JPA mappings with cascade and fetch strategies
- **Prerequisite System** - Self-referencing course prerequisite with validation
- **Real-time Search** - Client-side filtering with instant results
- **Modal-based Registration** - Clean UX with overlay and form validation
- **Status Management** - Enum-based status tracking (REGISTERED, COMPLETED, DROPPED)
- **Student Dashboard** - Comprehensive view with statistics and filtering
- **Responsive Design** - Mobile-friendly interface with adaptive layouts

6.2 Additional Enhancements

- **Custom Query Methods** - Derived queries in repositories for complex filtering
- **Data Seeding** - Automatic population of sample data on startup
- **Grade Tracking** - Optional grade field for completed courses
- **Registration Date** - Automatic timestamp on registration creation
- **Category Filtering** - Frontend categorization by course type
- **Loading States** - User feedback during asynchronous operations
- **Empty States** - Helpful messages when no data is available
- **Alert System** - Color-coded success/error notifications with auto-dismiss
- **Prerequisite Warnings** - Visual indicators for courses requiring prerequisites
- **Statistics Cards** - Real-time counts of registration statuses

6.3 Code Quality Practices

- **Clean Architecture** - Clear separation between layers (Controller → Service → Repository)
- **Single Responsibility** - Each class/component has one focused purpose
- **DRY Principle** - Reusable components and utility functions
- **Consistent Naming** - Standard conventions across backend and frontend
- **Error Handling** - Try-catch blocks with meaningful error messages
- **Type Safety** - Proper use of TypeScript-ready patterns
- **Code Comments** - Documentation for complex business logic
- **Git Best Practices** - Meaningful commit messages and branching

Challenges & Solutions

7.1 Technical Challenges

Challenge 1: Circular Dependency in Prerequisite Validation

Problem: Validating prerequisites required checking if a student has completed a course, which could lead to infinite loops in self-referencing relationships.

Solution: Implemented explicit null checks and status-based queries to break the circular reference chain. Used EXISTS query instead of fetching full entities.

Challenge 2: CORS Policy Blocking API Requests

Problem: React frontend on localhost:3000 was blocked from accessing Spring Boot backend on localhost:8080 due to CORS policy.

Solution: Added @CrossOrigin annotation to all controllers with explicit origin whitelisting. Configured proper CORS headers for preflight requests.

Challenge 3: Managing Bidirectional Relationships

Problem: JSON serialization caused infinite recursion with bidirectional @OneToMany relationships.

Solution: Used DTOs instead of directly serializing entities. Carefully applied @JsonManagedReference and @JsonBackReference where needed.

Challenge 4: Real-time Data Synchronization

Problem: After successful registration, course list didn't reflect updated seat availability.

Solution: Implemented data refresh mechanism that re-fetches course list after successful registration, ensuring UI stays in sync with backend state.

7.2 Design Decisions

Decision	Rationale
Enum for Registration Status	Type-safe, prevents invalid status values, easier to maintain

DTO Pattern for Registration	Decouples API from entity structure, allows flexible request/response formats
Service Layer for Business Logic	Keeps controllers thin, makes business rules reusable and testable
Custom Exception Classes	Provides semantic error handling, enables specific error messages
Modal for Registration Form	Reduces page navigation, provides focused user experience
Component-based Frontend	Promotes reusability, easier maintenance, clear separation of concerns

Conclusion

This project successfully demonstrates a complete full-stack application implementing course registration and enrollment management. The system effectively combines Spring Boot's powerful backend capabilities with React's modern frontend framework to create a user-friendly, robust application.

Learning Outcomes Achieved:

- **JPA & Hibernate Mastery** - Complex entity relationships, custom queries, validation
- **RESTful API Design** - Proper HTTP methods, status codes, error handling
- **Business Logic Implementation** - Multi-layered validation with domain rules
- **Frontend-Backend Integration** - Seamless communication using Axios and REST
- **Modern React Development** - Hooks, component composition, state management
- **Error Handling Architecture** - Custom exceptions with centralized handling
- **UI/UX Design Principles** - Responsive, accessible, user-friendly interface
- **Full Development Lifecycle** - From requirements to testing to deployment

Technical Stack Summary:

Layer	Technologies
Backend Framework	Spring Boot 3.x, Spring Data JPA, Spring Web

Database	MySQL 8.0, Hibernate ORM
Frontend Framework	React 18, React Router v6
HTTP Client	Axios for API communication
Styling	Custom CSS with modern design system
Build Tools	Maven (Backend), npm/Create React App (Frontend)
Testing Tools	Postman for API testing

Project Metrics:

Metric	Count
Backend Classes	15+ (Entities, Controllers, Services, Repositories)
Frontend Components	4 major components + API utility
API Endpoints	20+ REST endpoints
Business Rules	3 comprehensive validation rules
Database Tables	3 entities with relationships
Lines of Code	2000+ (Backend + Frontend combined)

The project demonstrates not only technical proficiency but also attention to detail, code quality, and user experience. The clean architecture ensures maintainability, while the comprehensive error handling provides reliability. This system serves as a solid foundation that can be extended with additional features and scaled to handle enterprise-level requirements.

Thank you for reviewing this assignment!

This project was completed with dedication to best practices and modern development standards.

Course Registration System | Web Engineering (CSC-324) | Assignment 3

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